MODEL QUESTION PAPER

QP Code:……. **Reg. No. ……**
Name:…….

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FOURTH SEMESTER MGU – UGP (HONOURS) REGULAR

EXAMINATION MARCH 2026

Fourth Semester

Discipline Specific Elective – MG4DSECHE200 - Polymer Chemistry

(2024 Admission Onwards)

Duration: 2 Hrs. Maximum Marks: 70

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

[Learning Domain] [CO No(s)]

Part A

Very Short Answer Questions
Answer any ***ten*** questions.
Each questions carries 2 marks

1. Define synthetic polymers with one example. [U][1]
2. What type of bond links amino acids in proteins? [U][1]
3. What is free radical polymerisation? [U][1]
4. What is meant by bulk polymerisation? [U][2]
5. What is the reaction medium in emulsion polymerisation? [K][2]
6. What is number average molecular weight (M_n)? [U][2]
7. What is a polyolefin? Give two examples. [U][3]
8. Name one property of PVAc that makes it useful as an adhesive. [U][3]
9. Name the monomers used to prepare phenol–formaldehyde (PF) resin. [U][3]
10. What is a temperature – sensitive polymer? [U][4]
11. What is a conducting polymer? Give an example of a conducting polymer. [U][4]
12. Mention one application of poly(vinyl carbazole) in electronics. [U][4]

(10 x 2 = 20)

Part B

Short Answer Questions
Answer any ***six*** questions.
Each questions carries 5 marks

13. Mention two applications of self-healing polymers. [K][1]
14. Briefly explain the polymerisation mechanism is involved in Nylon-6 formation. [U][1]
15. What is Ziegler–Natta polymerisation? Name one alkene polymer produced using Ziegler–Natta catalysts. [U][2]
16. What is meant by tacticity in polymers? [U][2]
17. Describe two important applications of Nylon-6 and Nylon-6,6. [U][3]
18. Describe the activator system used in rubber vulcanisation. [U][3]
19. Why is divinylbenzene used as a cross-linking agent in polystyrene? [A][4]
20. Explain why Nomex is used in protective clothing. [A][4]

(6 x 5 = 30)

Part C

Short Essay Questions
Answer any ***two*** questions.
Each questions carries 10 marks

21. Explain the mechanism of condensation polymerisation with reference to the formation of polyesters. [U][1]
22. Apply the concept of chain flexibility with suitable examples and discuss the factors affecting chain flexibility. [A][2]
23. Briefly discuss the properties and applications of Kevlar and PET [U][3]
24. Explain how biodegradable polyurethanes function as drug release agents in controlled drug delivery systems. [A][4]

(2 x 10 = 20)

END OF THE QUESTION PAPER

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